



Cisco CCNA 1 summary

Netwerkarchitectuur 1 (Karel de Grote Hogeschool)



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Chapter 1: Explore the network

The ways we communicate

- Texting
- Social media
- Collaboration tools
- Blogs
- Wikis
- Podcasting
- Peer-to-Peer (P2P) file sharing

Sizes of networks

- PAN (Personal Area Network)
- LAN (Local Area Network) or WLAN (Wireless LAN)
- MAN (Metropolitan Area Network)
- WAN (Wide Area Network)
- SAN (Storage Area Network)

Clients and servers → all end devices called hosts

Peer-to-Peer → many computers function as the servers and clients on the network

Network components

- Devices (physical elements)
- Media
- Services

End devices

- Desktop Computer
- Laptop
- Printer
- IP phone
- Wireless Tablet
- TelePresence endpoint

Intermediary devices

- Wireless router
- LAN switch
- Router
- Multilayer switch
- Firewall appliance

Network media

- Copper
- Fiber optic
- Wireless

Network Interface Card (NIC) → provides the physical connection to the network

Physical Port → a connector or outlet on a networking device where the media is connected to an end device or another networking device

Interface → specialized ports on a networking device that connect to individual networks

Topology diagrams

- Physical topology diagrams → identify the physical location of intermediary devices and cable installation
- Logical topology diagrams → identify devices, ports, and addressing scheme

Tranets

- Intranet
- Extranet

Home and small office internet connections

- Cable
- DSL (Digital Subscriber Lines)
- Cellular
- Satellite
- Dial-up Telephone

Businesses internet connections

- Dedicated leased line
- Ethernet WAN
- DSL
- Satellite

Network architecture

- Fault tolerance
- Scalability
- Quality of Service (QoS)
- Security

New trends

- Bring your own device (BYOD)
- Online collaboration
- Video communications

- Cloud computing

Chapter 2: Configure a network operating system

Operating Systems (OS)

- Shell
- Kernel
- Hardware

Interfaces

- Command-line interface (CLI)
- Graphical user interface (GUI)

Access methods

- Console → the physical management port
- Secure Shell (SSH) → a remote secure CLI connection through a virtual interface
- Telnet → a remote insecure CLI connection through a virtual interface

Chapter 3: Network protocols and communications

Establishment rules

1. An identified sender and receiver
2. Common language and grammar
3. Speed and timing of delivery
4. Confirmation or acknowledgment requirements

Message timing

- Access Method → a method that determines when someone can send a message, when to begin sending messages
- Flow Control → control the time between 2 frames
- Response Timeout

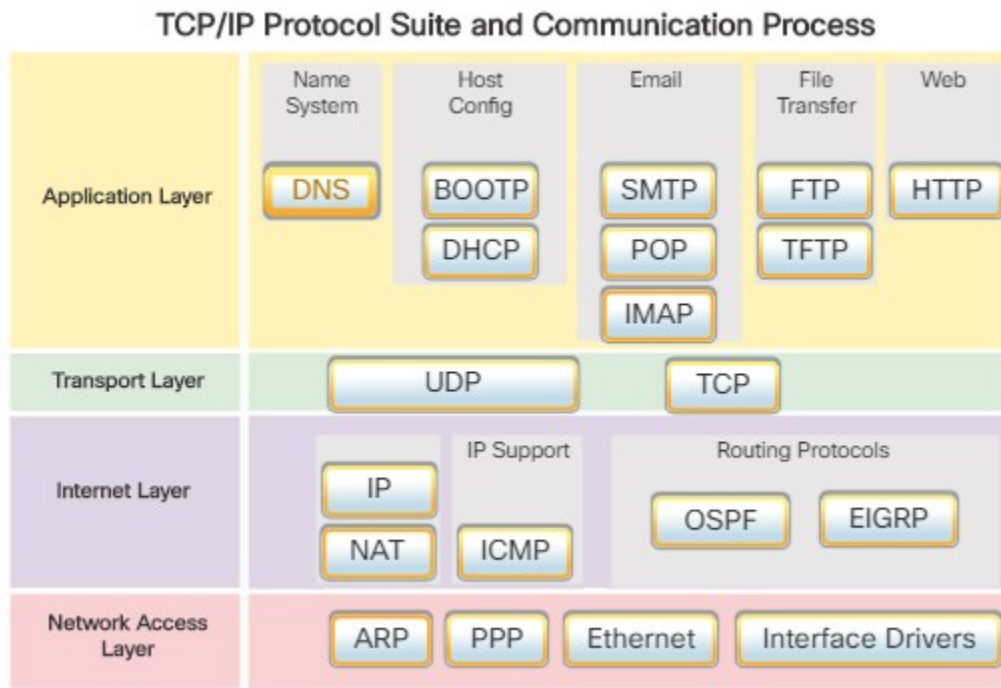
Message delivery options

- Unicast
- Multicast
- Broadcast

Protocol Suites and Industry Standards

Rectangular Snip

Layer Name	TCP/IP	ISO	AppleTalk	Novell Netware
Application	HTTP DNS DHCP FTP	ACSE ROSE TRSE SESE	AFP	NDS
Transport	TCP UDP	TP0 TP1 TP2 TP3 TP4	ATP AEP NBP RTMP	SPX
Internet	IPv4 IPv6 ICMPv4 ICMPv6	CONP/CMNS CLNP/CLNS	AARP	IPX
Network Access	Ethernet PPP Frame Relay ATM WLAN			

TCP/IP Protocol Suite

DNS → Domain Name System (or Service)

BOOTP → Bootstrap Protocol

DHCP → Dynamic Host Configuration Protocol

SMTP → Simple Mail Transfer Protocol

POP(3) → Post Office Protocol (version 3)

IMAP → Internet Message Access Protocol

FTP → File Transfer Protocol

TFTP → Trivial File Transfer Protocol

HTTP → Hypertext Transfer Protocol

UDP → User Datagram Protocol

TCP → Transmission Control Protocol

IP → Internet Protocol

NAT → Network Address Translation

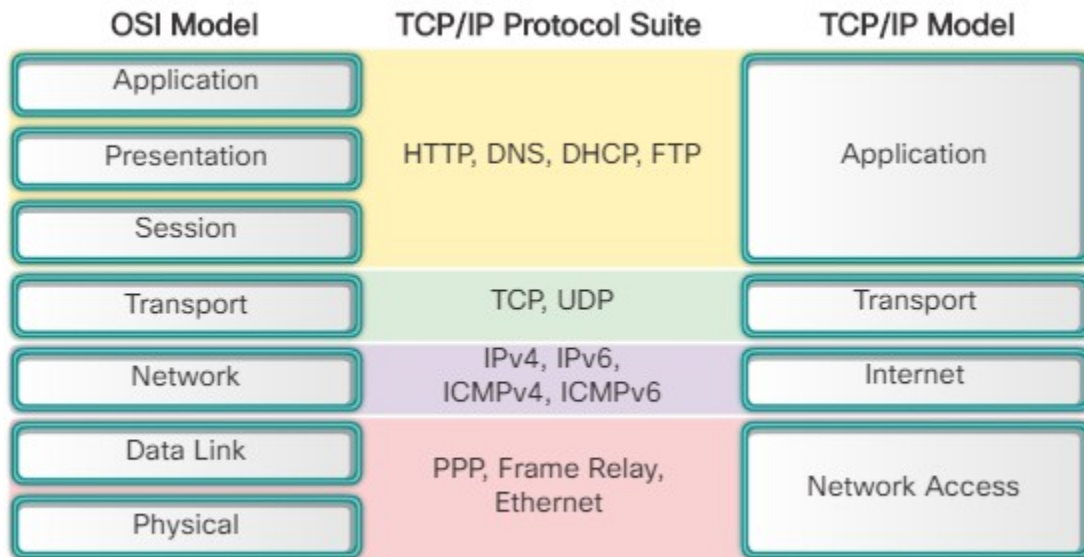
ICMP → Internet Control Message Protocol

OSPF → Open Shortest Path First

EIGRP → Enhanced Interior Gateway Routing Protocol

ARP → Address Resolution Protocol

PPP → Point-to-Point Protocol



The benefits of using a layered model

- Assisting in protocol design because protocols that operate at a specific layer have defined information that they act upon and a defined interface to the layers above and below
- Fostering competition because products from different vendors can work together
- Preventing technology or capability changes in one layer from affecting other layers above and below
- Providing a common language to describe networking functions and capabilities

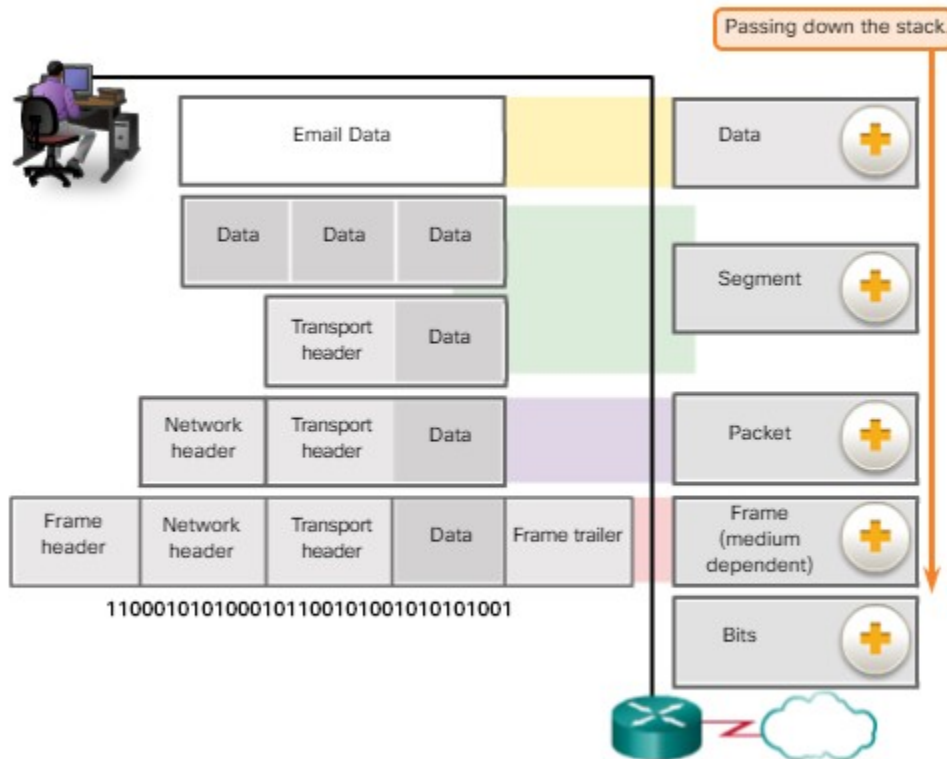
Protocol model → this type of model closely matches the structure of a particular protocol suite

Reference model → this type of model provides consistency within all types of networking protocols and services by describing what to be done at a particular layer

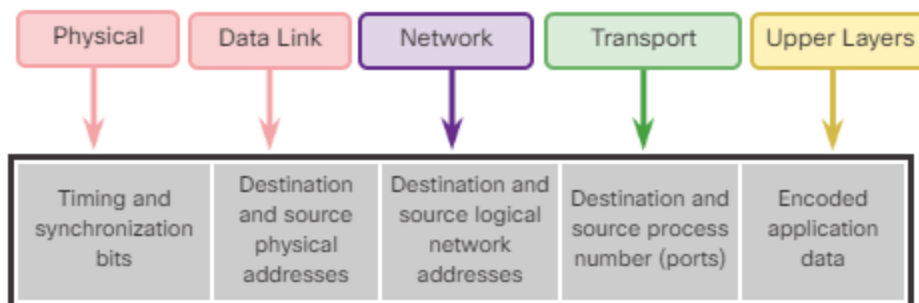
Segmentation → breaking communication into pieces

Multiplexing → multiple communications are interleaved, giving each user a part of the bandwidth

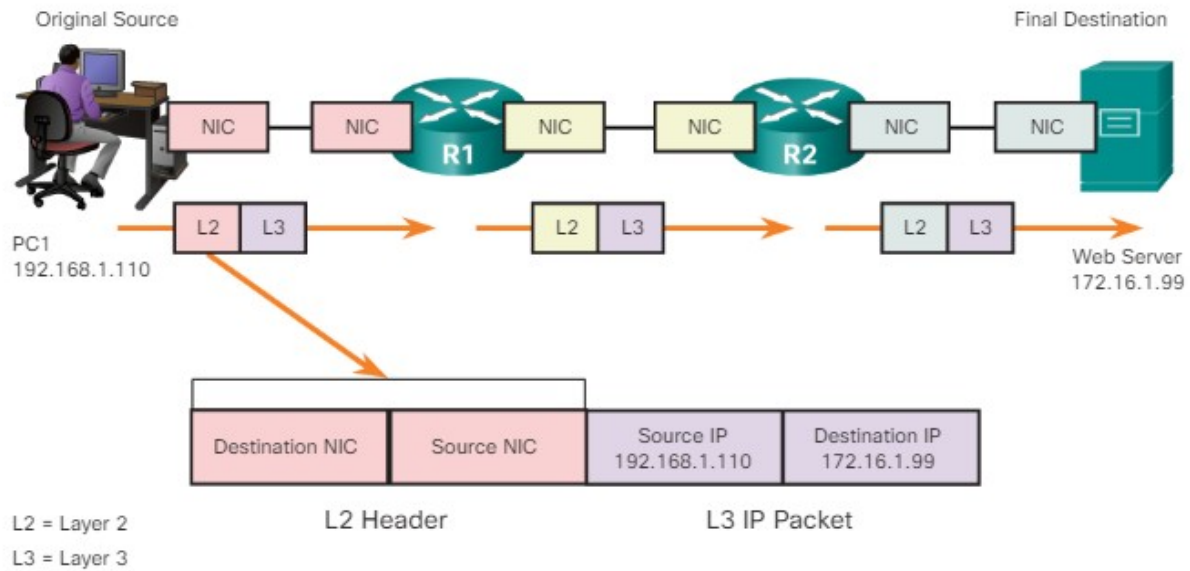
Encapsulation



Network addresses



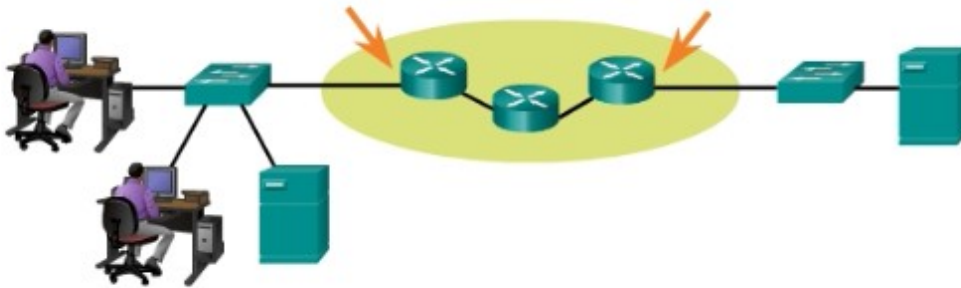
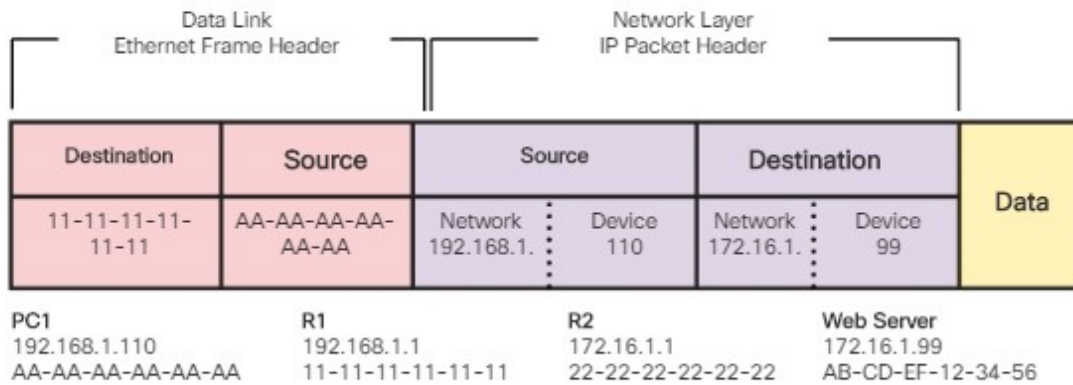
Data link addresses



Devices on the same network

- Network portion → the left-most part of the address that indicates which network the IP address is a member
- Host portion → the remaining part of the address that identifies a specific device on the network

Communicating with a Device on a Remote Network

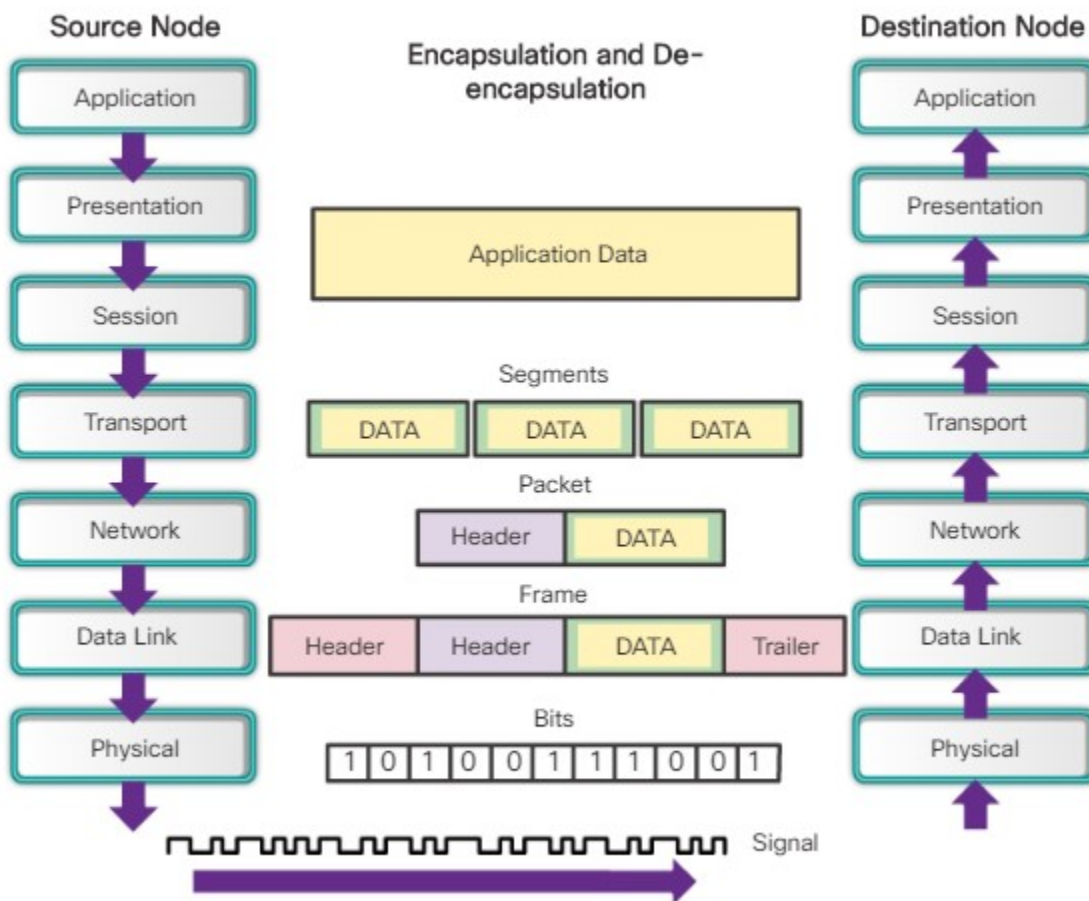


Chapter 4: Network access

Types of connections (home router)

- Ethernet Switch
- Internet Connection
- Embedded Wireless Antenna

The physical layer



Physical layer media

- Copper cable → the signals are patterns of electrical pulses
- Fiber-optic cable → the signals are patterns of light
- Wireless → the signals are patterns of microwave transmissions

Physical layer standards

- International Organization Standardization (ISO)
- Telecommunications Industry Association/Electronic Industries Association (TIA/EIA)

- International Telecommunication Union (ITU)
- American National Standards Institute (ANSI)
- Institute of Electrical and Electronics Engineers (IEEE)

Physical layer standards functions

1. Physical components → the electronic hardware devices, media, and other connectors that transmit and carry the signals
2. Encoding → is a method of converting a stream of data bits into a predefined “code”
3. Signaling → the physical layer must generate the electrical, optical, or wireless signals that represent the “1” and “0” on the media

Bandwidth → the rate of transfer of bits on a media

Types of physical media

- Fast Ethernet switch ports
- SHDSL interface
- Management ports
- Gigabit Ethernet interfaces
- USB type A connector

Interferences copper cable

- Electromagnetic interference (EMI) or radio frequency interference (RFI)
- Crosstalk → disturbance caused by the electric or magnetic fields of a signal on one wire in an adjacent wire

Copper media

- Unshielded Twisted-Pair (UTP)
- Shielded Twisted-Pair (STP)
- Coaxial

UTP cabling standards

- Cable types
- Cable lengths
- Connectors
- Cable termination
- Methods of testing cable

Types of UTP cable

- Ethernet Straight-through
- Ethernet Crossover
- Rollover

Types of fiber media

- Single-mode fiber (SMF) → laser
- Multimode fiber (MMF) → LED

Testing fiber cables

- Misalignment → the fiber-optic media aren't precisely aligned
- End gap → the media doesn't completely touch at the splice or connection
- End finish → the media ends aren't well polished, or dirt is present at the termination

Concerns of wireless media

- Coverage area
- Interference
- Security
- Shared medium (half-duplex)

Types of wireless media

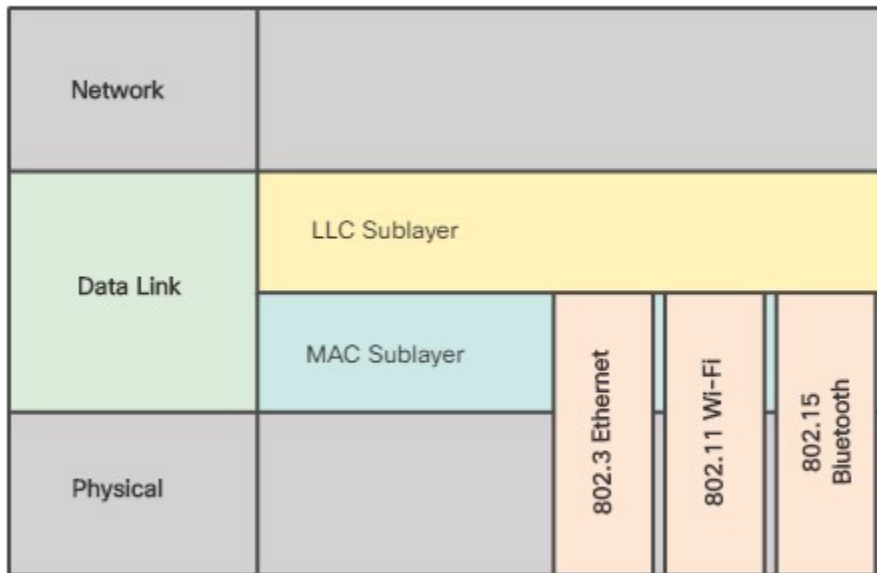
- Wi-Fi (standard IEEE 802.11)
- Bluetooth (standard IEEE 802.15)
- WiMAX (standard IEEE 802.16)

Responsibility data link layer

- Allowing the upper layers to access the media
- Accepting layer 3 packets and packaging them into frames
- Preparing network data for the physical network
- Controlling how data is placed and received on the media
- Exchanging frames between nodes over a physical network, such as UTP or fiber-optic
- Receiving and directing packets to an upper protocol
- Performing error detection

Data link sublayers

- Logical Link Control (LLC) → this upper sublayer communicates with the network layer, it placed information in the frame that identifies which network layer protocol is being used for the frame
- Media Access Control (MAC) → this lower sublayer defines the media access processes performed by the hardware, it provides data link layer addressing and access to various network technologies

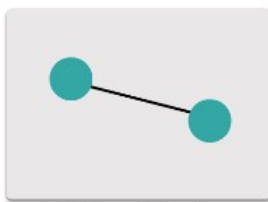


Topology → how the connection between the nodes appears to the data link layer

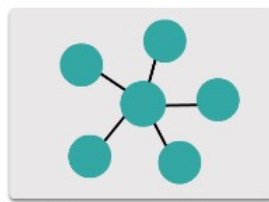
Media sharing → how the nodes share the media

Common physical WAN topologies

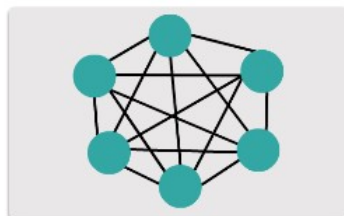
- Point-to-Point
- Hub and spoke
- Mesh



Point-to-point topology



Hub and spoke topology



Full mesh topology

Physical LAN topologies

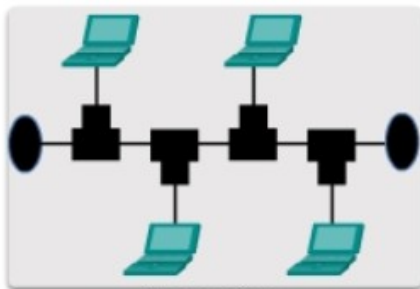
- Star
- Extended star
- Bus
- Ring



Star topology



Extended star topology



Bus topology

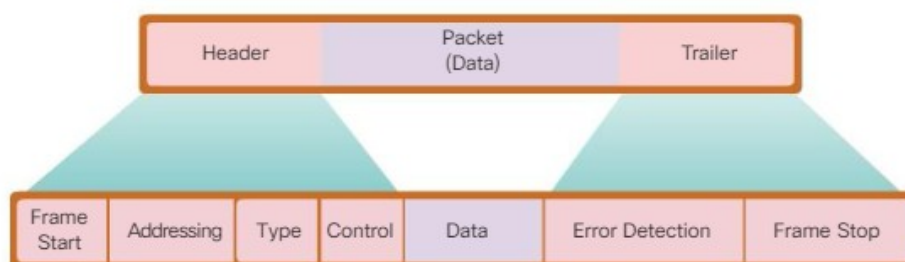


Ring topology

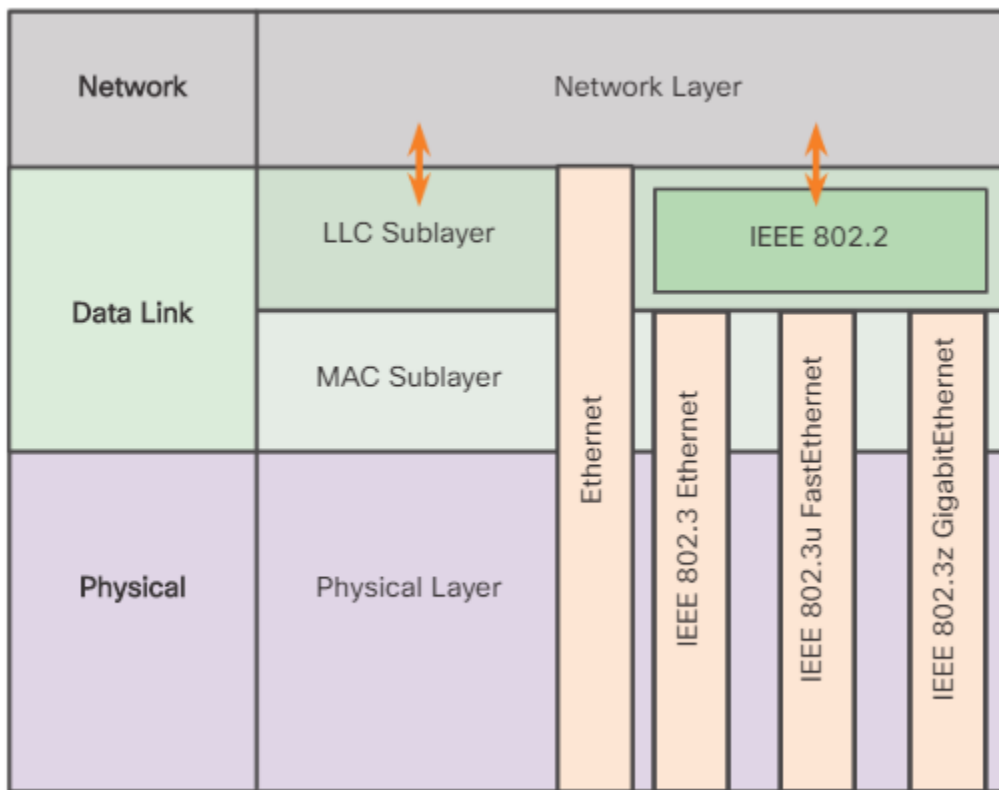
Duplex communication

- Half-duplex communication → both devices can transmit and receive on the media but not at the same time
- Full-duplex communication → both devices can transmit and receive on the media at the same time

Data link layer frame fields



Chapter 5: Ethernet



MAC sublayer

- Data encapsulation
 - Frame delimiting → the framing process provides important delimiters that are used to identify a group of bits that make up the frame
 - Addressing → the encapsulation process contains the layer 3 PDU and provides for data link layer addressing
 - Error detection → each frame contains a trailer used to detect any errors in the transmissions
- Media access control
 - Control of frame placement on and off the media
 - Media recovery

Broadcast MAC address → FF-FF-FF-FF-FF-FF

Multicast MAC address → MAC address start with "01-00-5E"

Frame forwarding methods (on cisco switches)

- Store-and-forward switching → receives the entire frame, and computes the CRC. If the CRC is valid, the switch looks up the destination address

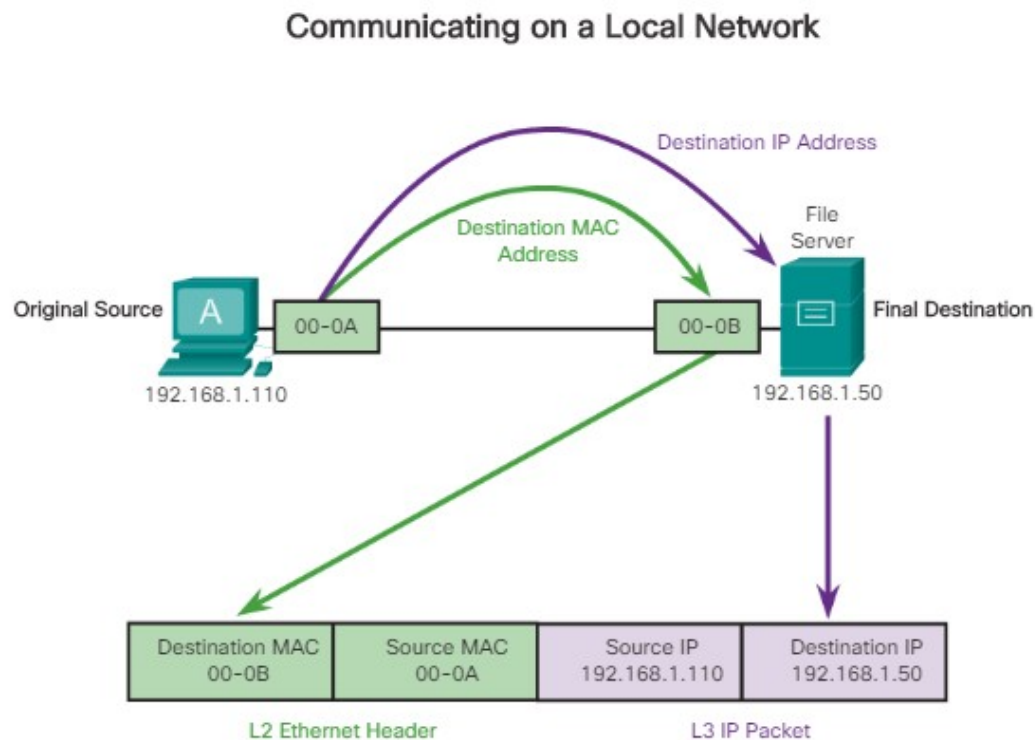
- Cut-through switching → forwards the frame before it is entirely received, the destination address of the frame must be read before the frame can be forwarded
 - Fast-forward switching → offers the lowest level of latency, fast-forwarding switching immediately forwards the packet after reading the destination address
 - Fragment-free switching → the switch stores the first 64 bytes of the frame before forwarding, the fragment-free switching can be viewed as a compromise between store-and-forward switching and fast-forward switching

Memory buffering on switches

- Port-based memory buffering → in port-based memory buffering, frames are stored in queues are linked to specific incoming and outgoing ports
- Shared memory buffering → shared memory buffering deposits all frames into a common memory buffer, which all the ports on the switch share

Auto-MDIX → detects the type of connection required and configures the interface accordingly (the auto-MDIX is disabled by default)

Destination on same network



- Physical address (the MAC address) → used for Ethernet NIC to Ethernet NIC communications on the same network
 - Destination MAC address → this is the MAC address of the next destination (continue changing)

- Source MAC address → this is the MAC address of the source (continue changing)
- Logical address (the IP address) → used to send the packet from the original source to the final destination
 - Source IP address → this is the IP address of the original source
 - Destination IP address → this is the IP address of the final destination

Functions of ARP

- Resolving IPv4 addresses to MAC addresses
 - If the packet's destination IPv4 address is on the same network as the source IPv4 address, the device will search the ARP table for the destination IPv4 address
 - If the destination IPv4 address is on a different network than the source IPv4 address, the device will search the ARP table for the IPv4 address of the default gateway
- Maintaining a table of mappings

ARP request

- Messages
 - Target IPv4 address → this is the IPv4 address that requires a corresponding MAC address
 - Target MAC address → this is the unknown MAC address and will be empty in the ARP request message
- Header
 - Destination MAC address → this is a broadcast address requiring all Ethernet NIC's on the LAN to accept and process the ARP request
 - Source MAC address → this is the sender of the ARP request's MAC address
 - Type → ARP messages have a type field of 0x806, this informs the receiving NIC that the data portion of the frame needs to be passed the ARP process

ARP reply

- Messages
 - Sender's IPv4 address → this is the IPv4 address of the sender, the device whose MAC address was requested
 - Sender's MAC address → this is the MAC address of the sender, the MAC address needed by the sender of the ARP request
- Header
 - Destination MAC address → this is the MAC address of the sender of the ARP request
 - Source MAC address → this is the MAC address of the sender of the ARP reply's MAC address
 - Type → ARP messages have a type field of 0x806, this informs the receiving NIC that the data portion of the frame needs to be passed to the ARP process

ARP spoofing → this is a technique used by an attacker to reply to an ARP request for an IPv4 address belonging to another device, such as the default gateway. The attacker sends an ARP reply with his own

MAC address, the receiver of the ARP reply will add the wrong MAC address to its ARP table and send these packets to the attacker

Chapter 6: Network layer

Processes of the network layer

- Addressing end devices → end devices must be configured with a unique IP address for identification on the network
- Encapsulation → the network layer encapsulates the protocol data unit (PDU) from the transport layer into a packet
- Routing → the network layer provides services to direct packets to a destination host on another network, to travel to other networks, the packet must be processed by a router
- De-encapsulation → when the packet arrives at the network layer of the destination host, the IP header will be removed

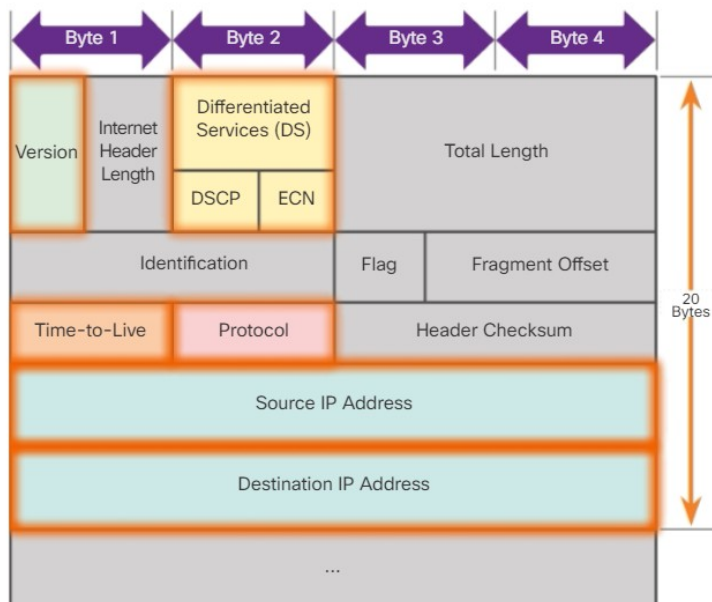
Network layer protocols

- Internet Protocol version 4 (IPv4)
- Internet Protocol version 6 (IPv6)

Characteristics of the IP protocol

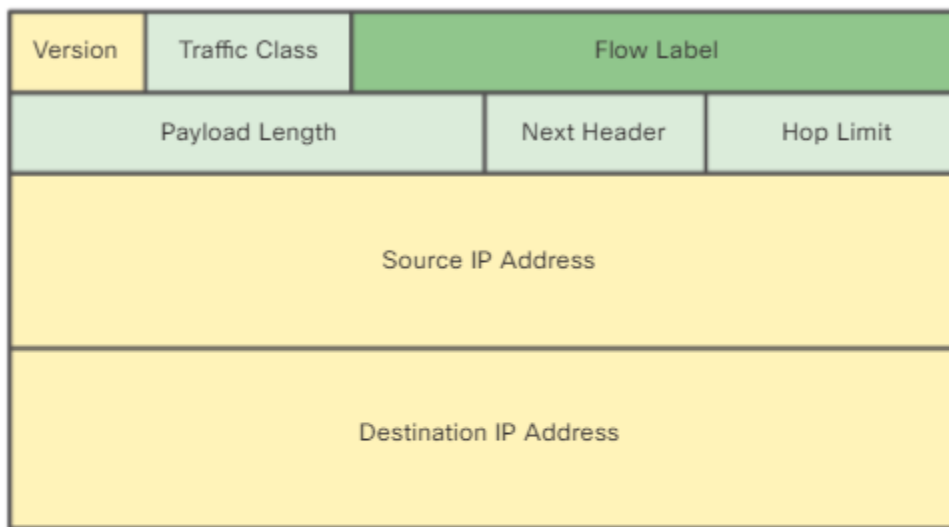
- Connectionless → no connection with the destination is established before sending data packets
 - The sender doesn't know: receiver is present, packet arrived, and receiver can read the packet
 - The receiver doesn't know when the packet is coming
- Best effort → delivery isn't guaranteed
- Media independent → operation is independent of the medium carrying the data

IPv4 packet



- Header
 - Version → contains a 4-bit binary set to '0100', identifies this is an IP version 4 packet
 - Differentiated Services (DS) → the DS field is an 8-bit field used to determine the priority of each packet (formerly called the Type of Service field)
 - Time-To-Live (TTL) → contains an 8-bit binary that is used to limit the lifetime of a packet, TTL decreased by one each time the packet is processed by a router
 - Protocol → this is an 8-bit binary value, that indicates the data payload type that the packet is carrying
 - Source IP address → contains a 32-bit binary value, that represents the source IP address of the packet
 - Destination IP address → contains a 32-bit binary value, that represents the destination IP address of the packet
- Limitations
 - IP address depletion → IPv4 has a limited number of unique public IPv4 addresses available
 - Internet routing table expansion → IPv4 routes consume a great deal of memory and processor resource on internet routes
 - Lack of end-to-end connectivity → Network Address Translation (NAT) is a technology implemented within IPv4 networks, NAT provides a way for multiple devices to share a single public IPv4 address. The IPv4 address of an internal network host is hidden, this can be problematic for technologies that require end-to-end connectivity

IPv6 packet

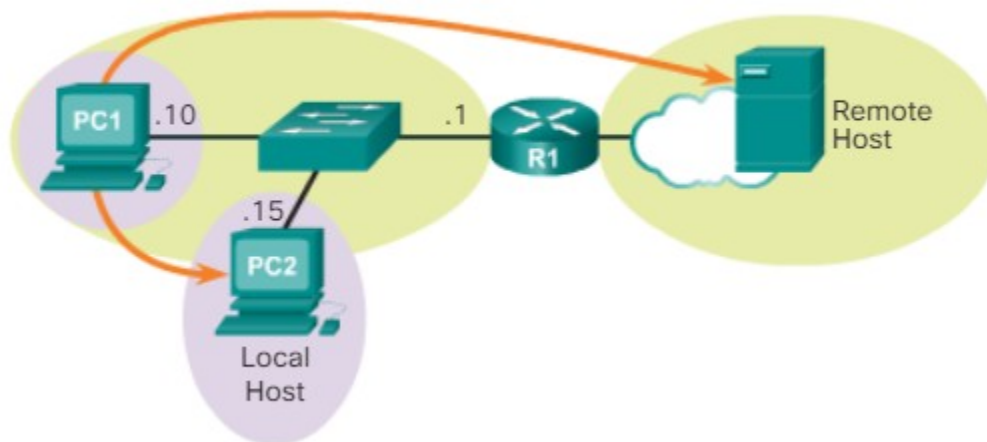


Legend

- Field names kept from IPv4 to IPv6
- Name and position changed in IPv6
- New field in IPv6

- Improvements
 - Increased address space → IPv6 addresses are based on 128-bit hierarchical addressing (340 undecillion addresses)
 - Improved packet handling → the IPv6 header has been simplified with fewer fields
 - Eliminates the need for NAT → with such a large number of public IPv6 addresses, this avoids some of the NAT-included application problems experienced by applications requiring end-to-end connectivity
- Header
 - Version → contains a 4-bit binary value set to '0110', that identifies this is an IP version 6 packet
 - Traffic class → this 8-bit field is equivalent to the IPv4 differentiated services field
 - Flow label → this 20-bit field suggest that all packets with the same flow label receive the same type of handling by routers
 - Payload length → this 16-bit field indicates the length of data portion or payload of the IPv6 packet
 - Next header → this 8-bit field is equivalent to the IPv4 protocol field
 - Hop limit → this 8-bit field replaces the IPv4 TTL field
 - Source address → this 128-bit field identifies the IPv6 address of the sending host
 - Destination address → this 128-bit field identifies the IPv6 address of the receiving host

Host forwarding decision



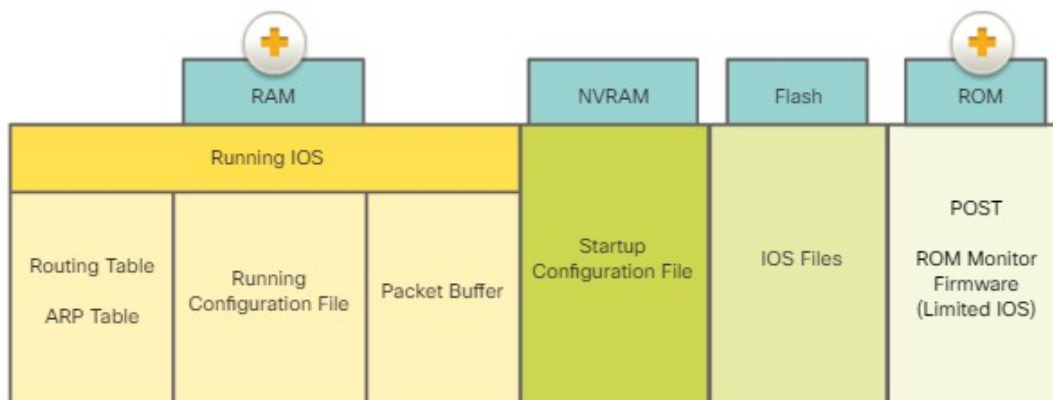
- Itself → a host can ping itself by sending a packet to a special address of '127.0.0.0', which is referred as the loopback interface
- Local host → this is a host on the same local network as the sending host
- Remote host → this is a host on a remote network, the hosts don't share the same network address

Default gateway → the default gateway is the network device that can route traffic to other networks, it is the router that can route traffic out of the local network

Router packet forwarding decision

- Directly-connected routes → these routes come from the active router interface
- Remote routes → these routes come from remote networks connected to other routers
- Default route → like a host, routers also use a default route as a last resort. If there is no other route to the desired network in the routing table

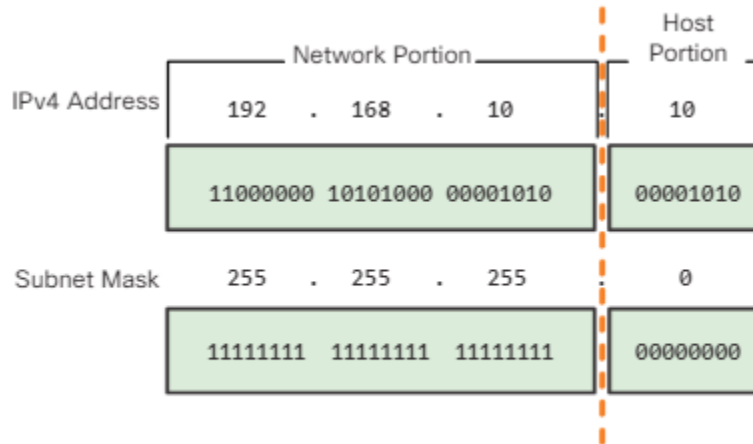
Router



- Types
 - Branch → teleworkers, small businesses, and medium-size branch sites
 - WAN → large businesses, organizations, and enterprises
 - Service provider → large service providers
- Memory
 - RAM → this is volatile memory used in Cisco routers to store applications, processes, and data needed to be executed by the CPU
 - ROM → this is non-volatile memory used to store crucial operational instructions and limited IOS
 - NVRAM → this memory is used as the permanent storage for the startup configuration file
 - Flash → flash memory is non-volatile computer memory used as permanent storage for the IOS and other system related files such as log files, voice configuration files, HTML files, backup configurations, and more

Chapter 7: IP addressing

IPv4 address structure



- IPv4 address → unique IPv4 address of the host
- Subnet mask → used to identify the network/host portion of the IPv4 address
- Default gateway → identifies the local gateway to reach remote networks

IPv4 prefix length

Subnet Mask	32-bit Address	Prefix Length
255.0.0.0	11111111.00000000.00000000.00000000	/8
255.255.0.0	11111111.11111111.00000000.00000000	/16
255.255.255.0	11111111.11111111.11111111.00000000	/24
255.255.255.128	11111111.11111111.11111111.10000000	/25
255.255.255.192	11111111.11111111.11111111.11000000	/26
255.255.255.224	11111111.11111111.11111111.11100000	/27
255.255.255.240	11111111.11111111.11111111.11110000	/28
255.255.255.248	11111111.11111111.11111111.11111000	/29
255.255.255.252	11111111.11111111.11111111.11111100	/30

IPv4 communication

- Unicast → the process of sending a packet from one host to an individual host
- Broadcast → the process of sending a packet from one host to all hosts in the network
- Multicast → the process of sending a packet from one host to a selected group of hosts, possibly in different networks (224.0.0.0 to 239.255.255.255 addresses are reserved for multicasting)

Special IPv4 addresses

- Private addresses
 - 10.0.0.0/8 to 10.255.255.255
 - 172.16.0.0/12 to 172.31.255.255
 - 192.168.0.0/16 to 192.168.255.255
- Loopback addresses
 - 127.0.0.0/8 to 127.255.255.254
- Link-Local addresses
 - 169.254.0.0/16 to 169.254.255.254
- TEST-NET addresses
 - 192.0.2.0/24 to 192.0.2.255

Classes IPv4 addresses

- Class A → 0.0.0.0/8 to 127.0.0.0/8
- Class B → 128.0.0.0/16 to 191.255.0.0/16
- Class C → 192.0.0.0/24 to 223.255.255.0/24
- Class D → 224.0.0.0 to 239.0.0.0
- Class E → 240.0.0.0 to 255.0.0.0

IPv4 and IPv6 coexistence

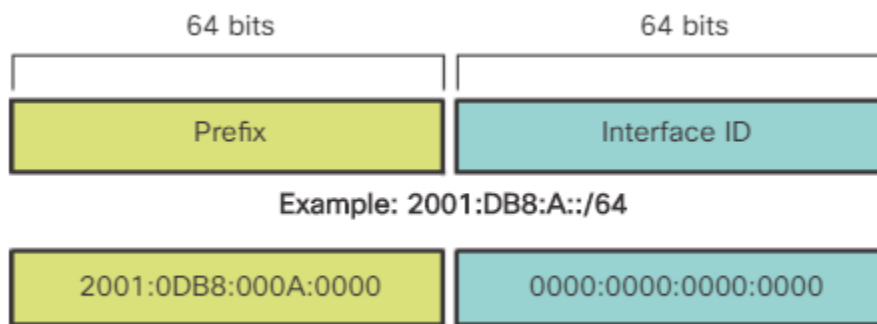
- Dual stack → dual stack allows IPv4 and IPv6 to coexist on the same network segment, dual stack devices run both IPv4 and IPv6 protocol stacks simultaneously
- Tunneling → tunneling is a method of transporting an IPv6 packet over an IPv4 network, the IPv6 packet is encapsulated inside an IPv4 packet
- Translation → Network translation allows IPv6 -enabled devices to communicate with IPv4-enabled devices using translation technique, an IPv6 packet is translated to an IPv4 packet and vice versa

IPv6 addressing

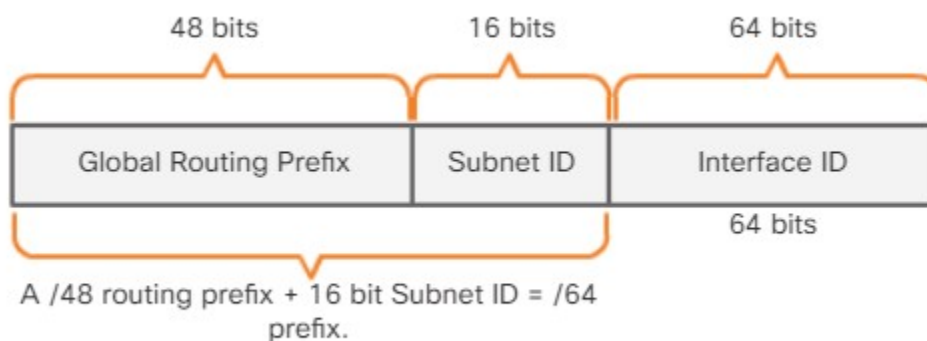
- Rules
 - Rule 1 → Omit leading 0s
 - Rule 2 → Omit all 0 segments
- Types
 - Unicast → an IPv6 unicast address uniquely identifies an interface on an IPv6-enabled device
 - Global unicast → a global unicast address is like a public IPv4 address, these are globally unique
 - Link-local → link-local addresses are used to communicate with other devices on the same local link
 - Unique local (FC00::/7 to FDFF::/7) → IPv6 unique local addresses are like private addresses for IPv4, but there are significant differences. Unique local addresses are used for local addressing within a site or between a limited number of sites, these addresses shouldn't be routable

- Loopback → ::1/128
- Unspecified address → ::/128
- Multicast → an IPv6 multicast address is used to send a single IPv6 packet to multiple destinations
 - All-nodes multicast group (FF02::1) → this is a multicast group that all IPv6-enabled devices join. A packet sent to this group is received and processed by all IPv6 interfaces on the link or network
 - All-routers multicast group (FF02::2) → this is a multicast group that all IPv6 routers join. A router becomes a member of this group when it is enabled as an IPv6 router with the ipv6 unicast-routing global configuration command
- Anycast → an IPv6 anycast address is any IPv6 unicast address that can be assigned to multiple devices

IPv6 prefix length



IPv6 global unicast



- Structure
 - Global routing prefix → the global routing prefix is the prefix, or network, portion of the address that is assigned by the provider
 - Subnet ID → the subnet ID is used by an organization to identify subnets within its site

- Interface ID → the IPv6 interface ID is equivalent to the host portion of an IPv4 address, the term interface ID is used because a single host may have multiple interface, each having one or more IPv6 addresses
- Dynamic configuration
 - RA option 1: SLAAC → SLAAC is stateless, which means there is no central server, router will create a prefix and interface ID (a random 64-bit number)
 - RA option 2: SLAAC and stateless DHCPv6
 - The router's link-local address, the RA's source IPv6 address for the default gateway address
 - A stateless DHCPv6 server to obtain other information such as a DNS server address and a domain name
 - RA option 3: stateful DHCPv6 → stateful is like DHCP for IPv4, a device can automatically receive its addressing information including a global unicast address, prefix length, and the addressing of DNS servers
 - The router's link-local address, the RA's source IPv6 address for the default gateway address
 - A stateful DHCPv6 server to obtain a global unicast address, DNS server address, domain name and all other information

Traceroute

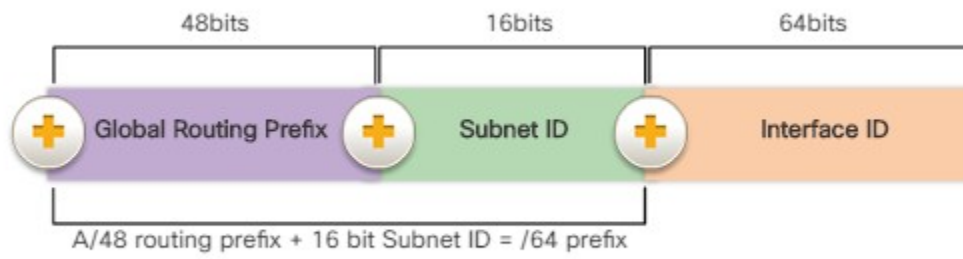
- Round Trip Time (RTT) → using traceroute provides round trip for each hop along the path and indicates if a hop fails to respond, the round trip time is the time a packet takes to reach the remote host and for the response from the host to return

Chapter 8: Subnetting IP Networks

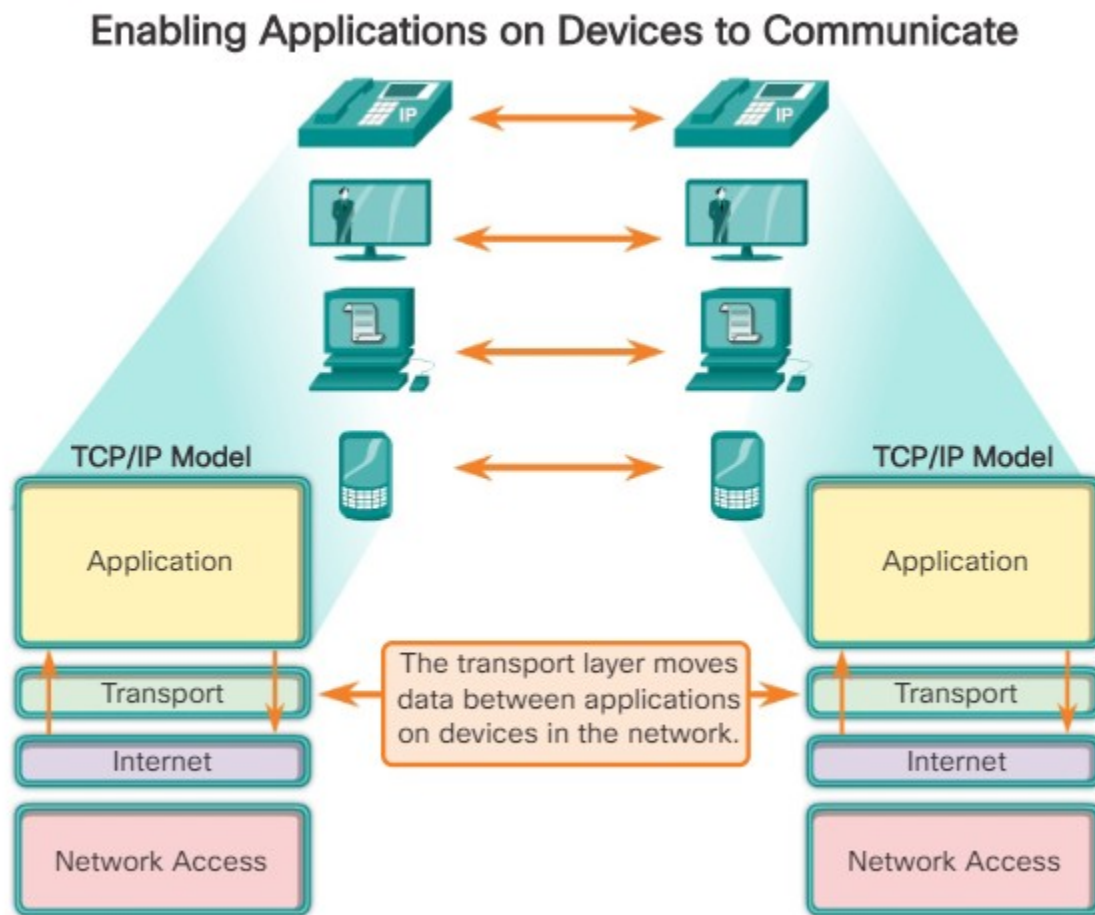
IPv4 subnetting

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of hosts
/8	255.0.0.0	nnnnnnnn . hhhhhhhh . hhhhhhhh . hhhhhhhh 11111111 . 00000000 . 00000000 . 00000000	16,777,214
/16	255.255.0.0	nnnnnnnn . nnnnnnnn . hhhhhhhh . hhhhhhhh 11111111 . 11111111 . 00000000 . 00000000	65,534
/24	255.255.255.0	nnnnnnnn . nnnnnnnn . nnnnnnnn . hhhhhhhh 11111111 . 11111111 . 11111111 . 00000000	254

Subnet Address (256 Possible Subnets)	Host Range (65,534 possible hosts per subnet)	Broadcast
<u>10.0.0.0/16</u>	<u>10.0.0.1</u> - <u>10.0.255.254</u>	<u>10.0.255.255</u>
<u>10.1.0.0/16</u>	<u>10.1.0.1</u> - <u>10.1.255.254</u>	<u>10.1.255.255</u>
<u>10.2.0.0/16</u>	<u>10.2.0.1</u> - <u>10.2.255.254</u>	<u>10.2.255.255</u>
<u>10.3.0.0/16</u>	<u>10.3.0.1</u> - <u>10.3.255.254</u>	<u>10.3.255.255</u>
<u>10.4.0.0/16</u>	<u>10.4.0.1</u> - <u>10.4.255.254</u>	<u>10.4.255.255</u>
<u>10.5.0.0/16</u>	<u>10.5.0.1</u> - <u>10.5.255.254</u>	<u>10.5.255.255</u>
<u>10.6.0.0/16</u>	<u>10.6.0.1</u> - <u>10.6.255.254</u>	<u>10.6.255.255</u>

IPv6 subnetting

Chapter 9: Transport Layer



Transport layer responsibilities

- Tracking individual conversations → a host may have multiple applications that are communicating across the network simultaneously, each of these applications communicates with one or more applications on one or more remote hosts. It is the responsibility of the transport layer to maintain and track these multiple conversations
- Segmenting data and reassembling segments → data must be prepared to be sent across the media in manageable pieces, most networks have a limitation on the amount of data that can be included in a single packet. At the destination, the transport layer must be able to reconstruct the pieces of data into a complete data stream that is useful to the application layer
- Identifying the applications → the transport layer must identify the target application to pass data streams to the proper applications

Multiplexing → multiple applications can use the network at the same time

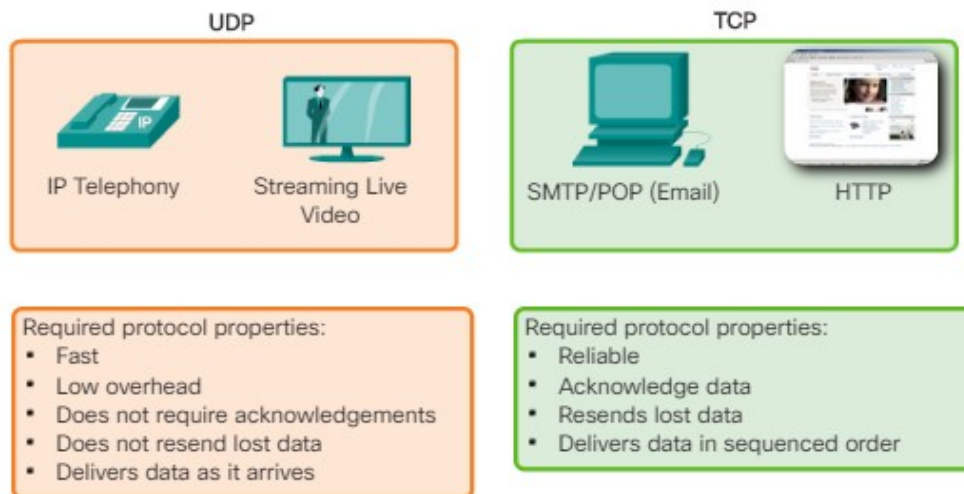
Segmentation → facilitates data transport by the lowest network layer

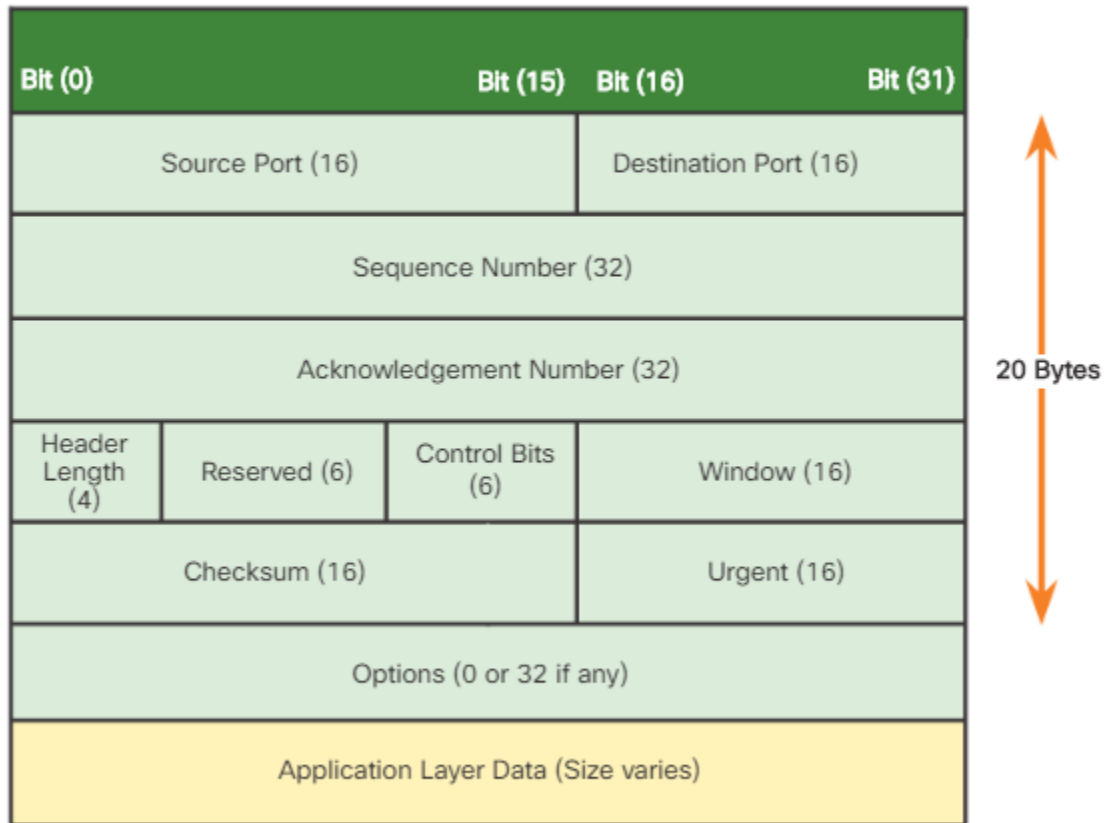
Error checking → error checking can be performed on the data in the segment to check if the segment was changed during the transmission

Transportation of Data

Transport Layer Protocols

Application developers choose the appropriate transport layer protocol based on the nature of the application.

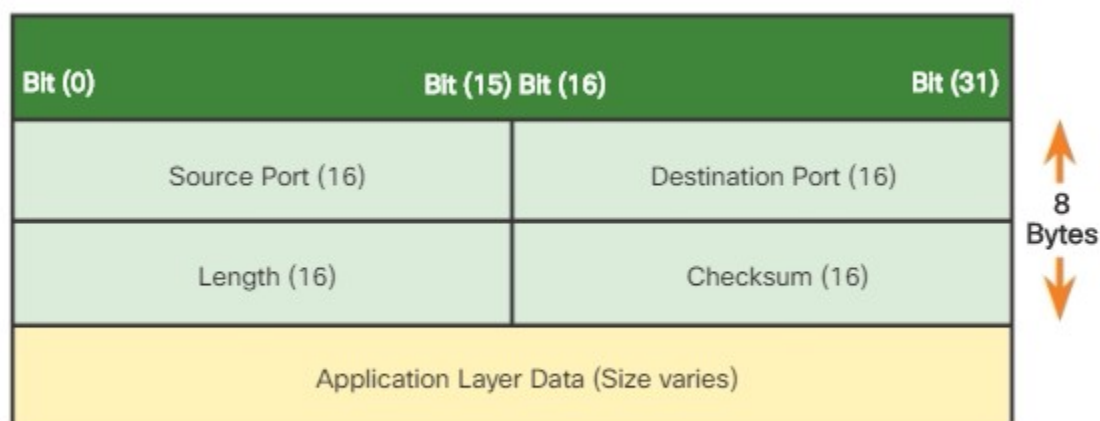


TCP**TCP Segment**

- **Features**
 - Establishing a session → TCP is a connection-oriented protocol, a connection-oriented protocol is one that negotiates and established a permanent connection (or session) between source and destination
 - Reliable delivery → ensuring that each segment that the source sends arrives at the destination
 - Same-Order delivery → ensures that the segments are reassembled into the proper order
 - Flow control → ensures that the receiver can process the data received
- **Header** → TCP is a stateful protocol
 - Source port (16 bits) and destination port (16 bits) → used to identify the application
 - Sequence number (32 bits) → used for data reassembly purposes
 - Acknowledgment number (32 bits) → indicates the data that has been received
 - Header length (4 bits) → known as “data offset”, indicates the length of the TCP segment header
 - Reserved (6 bits) → this field is reserved for the future

- Control bits (6 bits) → includes bit codes, or flags, which indicate the purpose and function of the TCP segment
- Window size (16 bits) → indicates the number of bytes that can be accepted at one time
- Checksum (16 bits) → used for error checking of the segment header and data
- Urgent (16 bits) → indicates if data is urgent
- Connection establishment (3 steps)
 - Step 1 → the initiating client request a client-to-server communication session with the server (source send SYN)
 - Step 2 → the server acknowledgment the client-to-server communication session and request a server-to-client communication session (destination receive SYN, destination send SYN and ACK)
 - Step 3 → the initiating client acknowledgment the server-to-client communication session
- Session termination (4 steps)
 - Step 1 → when the client has no more data to send in the stream, it sends a segment with the 'FIN' flag set
 - Step 2 → the server sends an 'ACK' to acknowledge the receipt of the 'FIN' to terminate the session from client-to-server
 - Step 3 → the server sends a 'FIN' to the client to terminate the server-to-client session
 - Step 4 → the client responds with an 'ACK' to acknowledge the 'FIN' from the server
- Applications
 - HTTP
 - FTP
 - SMTP
 - Telnet

UDP



- Features
 - No ordered data reconstruction → data is reconstructed in the order that it is received

- Connectionless → no session establishment
- Unreliable delivery → any segments lost are not resent
- No flow control → doesn't inform the sender about resource availability
- Header → UDP is a stateless protocol
 - Source port (16 bits)
 - Destination port (16 bits)
 - Length (16 bits)
 - Checksum (16 bits)
- Applications
 - Live video and multimedia applications → can tolerate some data lose, but require little or no delay (Ex. VoIP and live streaming video and IPTV)
 - Simple request and reply applications → applications with simple transactions where a host sends a request and may or not receive a reply (Ex. DNS and DHCP)
 - Applications that handle reliability themselves → unidirectional communications where the flow control, error detection, acknowledgment, and error recovery isn't required or can be handled by the application (Ex. SNMP and TFTP)

Port numbers

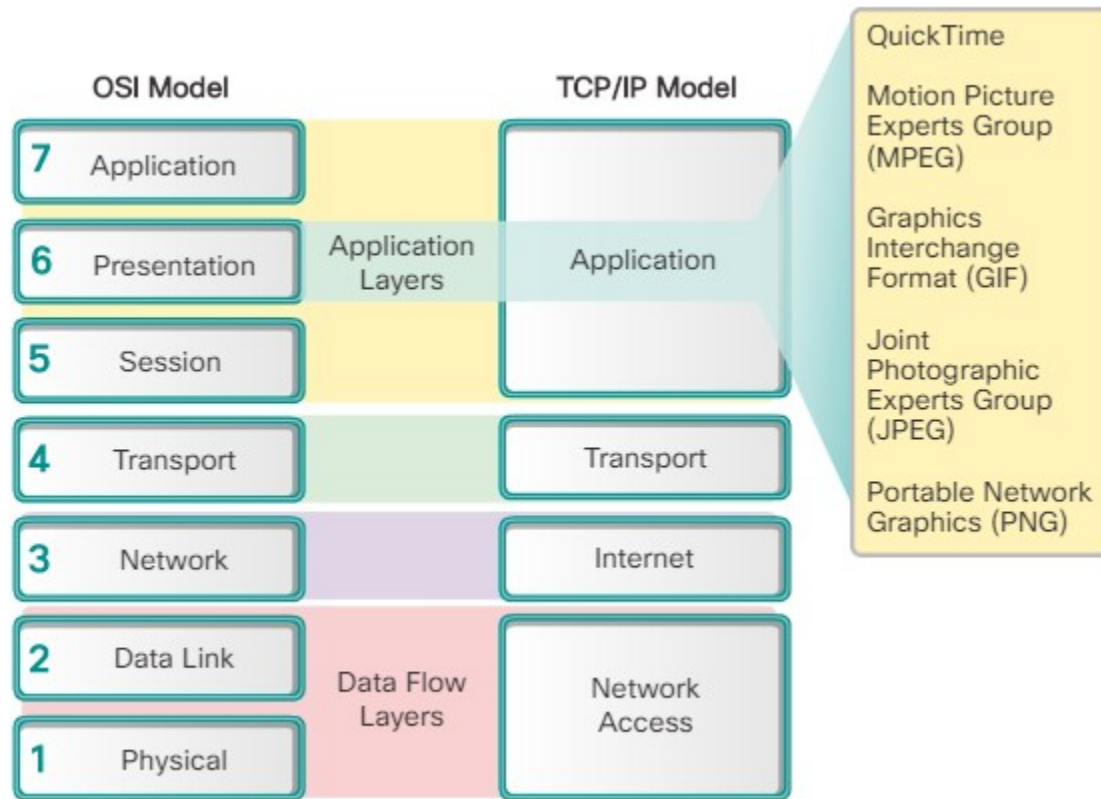
Port Number Range	Port Group
0 to 1023	Well-known Ports
1024 to 49151	Registered Ports
49152 to 65535	Private and/or Dynamic Ports

- Types
 - Source port → the source port number is dynamically generated by the sending device to identify a conversation between two devices
 - Destination port → the client places a destination port number in the segment to tell the destination server what services is being requested
- Groups
 - Well-know Ports (numbers 0 to 1023) → these numbers are reserved for services and applications, they are commonly used
 - Registered Ports (1024 to 49151) → these port numbers are assigned by IANA to a requesting entity to use with specific processes or applications, these processes are primarily individual applications that a user has chosen to install
 - Dynamic or Private Ports (numbers 49152 to 65535) → also know as ephemeral ports, they are usually assigned dynamically by the client's OS when a connection to a server is initiated

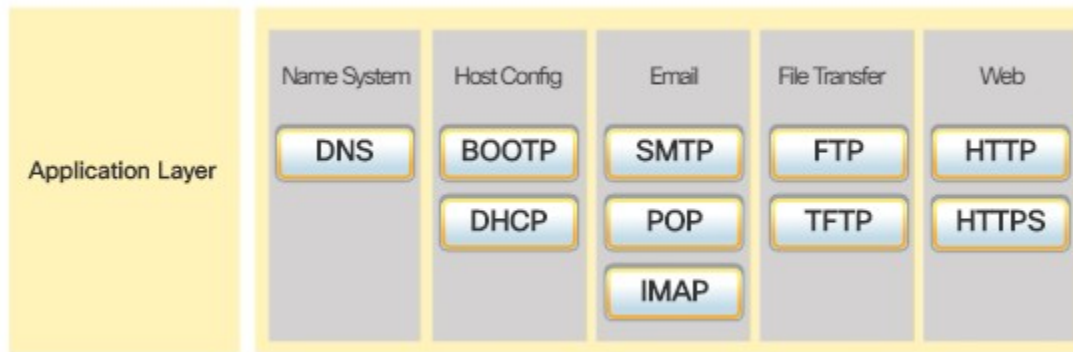
Socket Pairs → the combination of the source IP address and the source port number, or the destination IP address and destination port number

Chapter 10: Application Layer

Application layer protocols



- Application layer → the closest layer to the end user, the application layer provides the interface between the applications used to communicate and the underlying network over which messages are transmitted. Application protocols are used to exchange data between programs running on the source and the destination hosts
- Presentation layer (3 primary functions)
 - Formatting, or presenting, data at the source into a compatible form for receipt by the destination device
 - Compressing data in a way that can be decompressed by the destination device
 - Encrypting data for transmission and decrypting data upon receipt
- Session layer → the session layer creates and maintain dialogs between source and destination applications, the session layer handles the exchanges of information to initiate dialogs, keep them active, and to restart sessions that are disrupted or idle for a long period of time

TCP/IP application layer protocols

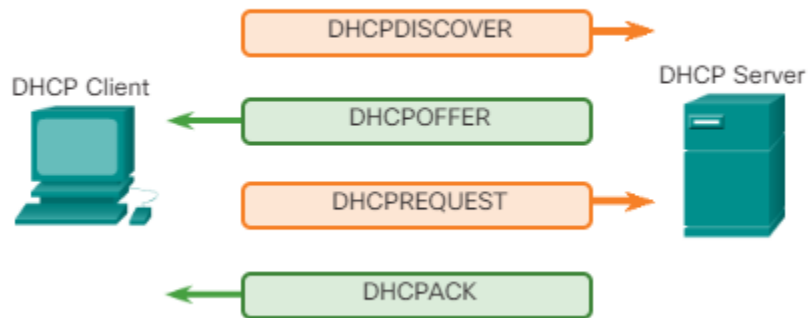
- Name System
 - DNS (Domain Name System -or Service)
- Host Config
 - BOOTP (Bootstrap Protocol)
 - DHCP (Dynamic Host Configuration Protocol)
- Email
 - SMTP (Simple Mail Transfer Protocol)
 - POP (Post Office Protocol)
 - IMAP (Internet Message Access Protocol)
- File Transfer
 - FTP (File Transfer Protocol)
 - TFTP (Trivial File Transfer Protocol)
- Web
 - HTTP (Hypertext Transfer Protocol)
 - HTTPS (Hypertext Transfer Protocol Secure)

HTTP and HTTPS message types

- GET → a client request for data, a client (web browser) sends the GET message to the web server to request HTML pages
- POST → uploads data files to the web server such as form data
- PUT → uploads resources or content to the web server such as an image

DNS hierarchy

- .com → a business or industry
- .org → a non-profit organization
- .au → Australian
- .co → Colombia
- ...

DHCP operation

1. DHCPDISCOVER
2. DHCPOFFER
3. DHCPREQUEST
4. DHCPACK

FTP process

1. Control Connection → client opens first connection to the server for control traffic (TCP port 21)
2. Data connection → client opens second connection for data traffic (TCP port 20)